# National University of Computer and Emerging Sciences

**Technical & Business Writing (SS2012)**
Date: 10th, April, 2026
**Course Instructors**
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**Sessional-II Exam**

**Total Time: 1 Hour**
**Total Marks: 30**
**Total Questions**: **03**

**Semester:** SP-2026
**Campus:** Karachi
**Dept:** CS

| Student Name | Roll No | Section | Student Signature |
|---|---|---|---|

***CLO #2: Compose reports for effective performance in professional life.***

**Question 1-A:** You are a 3rd-year Computer Science student who recently volunteered in your university's Procom 2026. During this event, you worked as part of a team responsible for managing a coding competition. **Now, write an 'experience' section entry for your resume.** **[4 marks]**

- **Write 3 bullet points**
- **Highlight at least 3 transferrable skills.**
- **Convert one of your bullet points into a quantified statement.**

**Procom 2026 – Coding Competition Volunteer**

- Coordinated with a team to manage the execution of a coding competition, demonstrating strong **teamwork** and **organizational skills**. [1 mark]
- Assisted participants by resolving technical queries and ensuring smooth system functionality, showcasing **problem-solving** and **communication skills**. [1 mark]
- Managed registration and monitored progress for over **120+ participants**, ensuring timely updates and efficient event flow, reflecting **time management** and **attention to detail**. [1.5 marks]

**Presentation of information 0.5 mark**
**Writing transferrable skills 2.5 marks**
**Quantified statement 1 mark**

**Question 1-B: Write a reference section and provide details of two referees. Follow the professional style with complete content.** **[4 marks]**

1. **Ms. Nazia Imam** **0.5**
   **Lecturer** **0.5**
   **FAST NUCES** **0.5**
   **Nazia.imam@nu.edu.pk** **0.5**
2. **Ms. Javeria Ali**
   **Lecturer**
   **FAST NUCES**
   **Javeria.ali@nu.edu.pk**

**Question 1-C: Select the correct option and write only the option number in your copy: [1 mark]**
**1. The main body of a cover letter should focus on:**

a) Detailing personal hobbies  b) Describing why the candidate is a good fit for the position
c) Listing every important job ever held
d) Asking when an interview can be scheduled

**2. What is the best way to close a cover letter?**

a) Thanks, talk soon!  b) Looking forward to your call
c) I appreciate your time and consideration. I look forward to the opportunity to discuss my qualifications further.
d) I look forward to discussing my qualifications and how they align with your needs. Let's schedule a time to connect.

**Question 2-A: write a complete set of instructions for sharing a compressed file through a cloud storage.**

**Title: 1 mark**
**Instructions: Imperative sentences + reference to visuals/visuals (3+1 marks)**
**Appropriate division of Steps+ logical flow and clarity = (3 marks)**

**[8 marks]**

***CLO #3: Design, develop, and document a research project.***

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**Question 3-A: Read the given abstract and answer the questions:**

i. **Write two purpose statements?** **[2 marks]**

This study aims to analyze the performance of cloud computing services for scientific computing workloads.

This research seeks to evaluate and compare the performance and cost models of commercial clouds with other scientific computing platforms.

ii. **Identify and write the problem statement in your own words?** **[4 marks]**

Cloud computing offers cost-effective, scalable resources but is built for web and database workloads. It remains unclear whether commercial clouds can meet the performance demands of scientific computing, especially Many-Task Computing workloads, due to virtualization and resource sharing penalties. Evaluating cloud performance against traditional platforms is essential to identify required improvements so clouds can become viable for the scientific community.

iii. **What is the key finding of the paper?** **[2 marks]**

The study finds that current cloud systems require significant (up to an order of magnitude) performance improvement to effectively support scientific computing workloads.

Cloud computing is an emerging commercial infrastructure paradigm that promises to eliminate the need for maintaining expensive computing facilities by companies and institutes alike. Through the use of virtualization and resource time-sharing, clouds serve with a single set of physical resources a large user base with different needs. Thus, clouds have the potential to provide to their owners the benefits of an economy of scale and, at the same time, become an alternative for scientists to clusters, grids, and parallel production environments. However, the current commercial clouds have been built to support web and small database workloads, which are very different from typical scientific computing workloads. Moreover, the use of virtualization and resource time-sharing may introduce significant performance penalties for the demanding scientific computing workloads. In this work we analyze the performance of cloud computing services for scientific computing workloads. We quantify the presence in real scientific computing workloads of Many-Task Computing (MTC) users, that is, of users who employ loosely coupled applications comprising many tasks to achieve their scientific goals. Then, we perform an empirical evaluation of the performance off our commercial cloud computing services including Amazon EC2, which is currently the largest commercial cloud. Last, wecomparethroughtrace-basedsimulationtheperformancecharacteristics and cost models of clouds and other scientific computing platforms, for general and MTC-based scientific computing work loads. Our results indicate that the current clouds need an order of magnitude in performance improvement to be useful to the scientific community, and show which improvements should be considered first to address this discrepancy between offer and demand.

**Question 3-B: Select the correct option and write ONLY the option number in your copy: [5 marks]**

**1. A paragraph states: "Despite advancements, students struggle to use AI tools effectively." This represents:**

a) Background c) Problem statement
b) Motivation d) Scope

**2. A sentence reads: "This research focuses only on undergraduate CS students in Karachi." This is:**

a) Scope c) Background
b) Problem d) Motivation

**3. A student explains: "This research is important because students increasingly rely on AI tools." This is:**

a) Background c) Scope
b) Motivation d) Method

**4.** A student includes statistics in the opening paragraph. This supports:

a) Motivation c) Scope
b) Background d) Conclusion

**5. A student writes: "This study only considers first-year students." This reflects:**

a) Purpose c) Problem
b) Scope d) Motivation

**6. Which is the correct order in introduction? [1 mark]**

a) Problem → Background → Purpose
b) Background → Problem → Purpose
c) Purpose → Background → Problem
d) Scope → Problem → Background

**7. Which element answers "why is this study needed?"**

a) Scope c) Method
b) Motivation d) Background

**8. A student includes citations in introduction. This supports:**

a) Credibility c) Method
b) Scope d) None

**9. Which element answers "what is included/excluded?"**

a) Scope c) Problem
b) Background d) Purpose